

EECS 841: Computer Vision

3 Credit Hours – Spring 2025

- **Instructor:**
Sumaiya Shomaji,
Assistant Professor,
Dept. of EECS,
University of Kansas
Email: shomaji@ku.edu
Office: Eaton 2020 and Nichols 251.
- **Lecture Time and Location:** T/Th 11:00AM – 12:15 PM, MAL 2048
- **Course Summary:** The course includes the fundamental principles and methods for image processing and computer vision. This course aims to understand how to efficiently represent, process, and analyze image signals. The course aims to provide students with the scientific foundations needed to implement and apply techniques used to address image analysis-related problems. Here is a tentative list of the topics that we will cover in the lectures:
 1. Overview
 2. Introduction
 3. Fundamentals of Imaging
 4. Binary Image Processing
 5. Spatial Domain Filtering
 6. Frequency Domain Processing
 7. Edges and Features
 8. Image Compression
 9. Color
 10. Segmentation
 11. Model Fitting
 12. Classification
 13. Video processing
- **Textbooks:** There is no required textbook, but the book below will be followed as a reference:
 - Title: Image Processing and Analysis
 - Author: Stan Birchfield
 - Publication date, edition, and publisher: 2017, First Edition, CL Engineering
 - ISBN: 978-1285179520
- **Software:**
 - Python
- **Office hours for Dr. Shomaji:** Online (prior email confirmation preferred)
 - Monday (2:00 PM – 3:00 PM) and Thursday (2:00 PM – 3:00 PM)
 - Zoom: <https://kansas.zoom.us/j/85954731810>
 - Meeting ID: 859 5473 1810
 - Passcode: 897155

- **Course Website and Discussion Forum:**

- Lecture notes and announcements will be posted on the Canvas.

- **Grading-methods of evaluation**

Your final grade will be computed by using the weighted average on the items below:

- 4 Homeworks: 40% (each will have 10%)
 - 2 Midterms: 20% (each will have 10%)
 - 1 Final: 20%
 - 1 Project: 20% (group-wise submission)
 - Attendance: 0% (attendance is highly encouraged)

Note that all the items mentioned above will contribute towards the final grade hence dropping any items is never an option. In the event of a regrade request, the student has to request within the next 2 days, and the entire homework assignment, project, quiz, or exam will be evaluated as well as regraded if necessary.

- **Grading Scale**

Your final grade in the class will be computed by using the weighted average given above, and the following scale:

A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
93<=	[89-93]	[86-89]	[83-86]	[79- 83]	[76-79]	[73-76]	[69-73]	[66-69]	[63-66]	[59-63]	<59

- **Tentative dates of exams:**

- Midterm1: Feb 18, 2025 (Time: during class)
 - Midterm2: March 25, 2025 (Time: during class)
 - Final: May 13, 2025 (Time: 10:30 AM – 1:00 PM)

- **Make-up exam policy for finals:** <https://registrar.ku.edu/final-exam-policies>

- **Exam accommodation policy:** <http://access.ku.edu/exam-accommodations-0>

- **Honesty Policy:** Plagiarism and unauthorized assistance related to any submission of this course is strictly prohibited. Please follow the Ethical writing guideline: <https://writing.ku.edu/academic-integrity>

- **HW and Project submissions:** Assignment questions will be posted in Canvas. Students will submit their answers on Canvas too. No handwritten submission will be allowed. Only PDFs of typed assignments uploaded using Canvas will be considered for evaluation. For the assignments involving coding, students should submit the .py and .PDF files of the code along with the writeup.

- **Late Submission Policy:** Late submission is never encouraged. However, if you submit any time after the deadline (even by a minute), for each day's delay you will lose 15% of whatever points you are going to earn. No submission will be accepted if you delay more than 3 days from the original deadline.

- **Exam medium:** Paper-based exam. You will attend in person and write on paper. Students must attend the exam on the due date.

- **Final Grade:** Final grades will depend completely on the student's performance on the course items. It will be determined from the weighted total points you will earn at the end of the semester. There won't be any extra credit exam/assignment/curving/bonus points to adjust the total weighted score and its corresponding grade.
 - **Academic Integrity:**

Students are NOT allowed to collaborate on *homework* assignments as a group. Each student is expected to do the coding and write up the assignment by himself/herself. No program code and writeup should be shared with another student or downloaded/copied from another source. Students are expected to collaborate on the *project* with a specified number of students and should not work by himself or herself, therefore. The students in each group will work together on mathematics, problem-solving, algorithm design and analysis, computer programming, application, and reporting. Each student is expected to contribute to the effort. Each group should turn in a single write-up for their project. However, the project group should not discuss the project with students outside of their group. For example, no mathematics or code should be shared with other project teams. Students are permitted to consult outside published material for the project to generate ideas to some extent. If a student (on the homework assignment) or a group (on the project) consults a source other than the lecture notes or textbook, he/she or they, respectively, must cite the source. Failure to cite the source will be considered cheating. No collaboration of any kind is allowed on the quizzes or exams. Exams are always closed book. Any form of collaboration with other students, help-seeking online or lecture notes, etc. during the exam. will be considered cheating.
 - **Artificial Intelligence Tools:** The use of Artificial Intelligence tools (e.g., ChatGPT) for writing solutions (or assisting others in writing their solutions either directly or indirectly) to homework assignments, quizzes, or exams or writing the project report is not permitted and is considered cheating. If you are feeling unsure about an assignment, it is important to use the allowable resources available to you, such as instructor office hours and/or tutoring.
 - **Penalties:** No form of cheating in the course will not be tolerated. Anyone found cheating for the first time will receive a 0 on that work. A second cheating incident will result in an F grade for the course. These penalties will be received by all parties involved, following a hearing with the instructor. In all cases, reports of academic misconduct will also be made to the Dean's office where further disciplinary action may be taken in accordance with School of Engineering and University of Kansas guidelines. This may result in much more serious sanctions. It is your responsibility not to let anyone copy your homework/project/quiz/exam; otherwise, you may have to pay the price for others' misconduct.
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